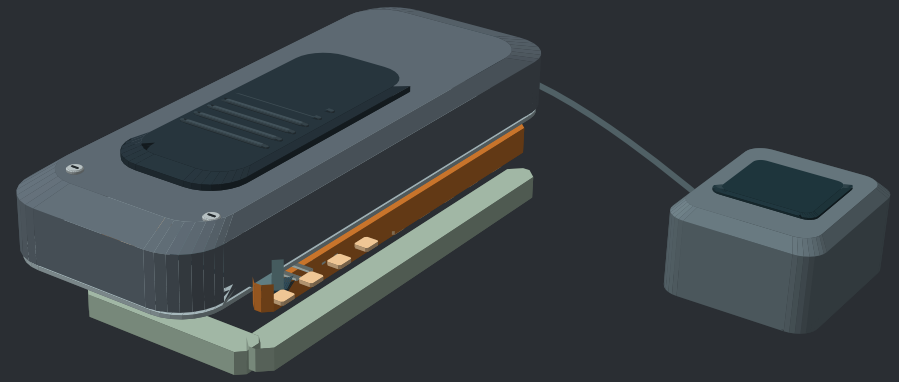
# VASCOUT

## Research discussion: FUJIFILM Sonosite



ILLUSTRATIVE PROBE + SEPARATE ACQUISITION UNIT

A proposed ultrasound-led vascular imaging system for clinician-controlled assessment. Tailored research prospect briefing.

**PROSPECT ONLY / INTEREST UNCONFIRMED / NOT SENT**

**Explore the live concept and evidence**

# Difficult access deserves clearer evidence.

Research intent: test whether anatomy and image-quality feedback can support clinician assessment.

CLINICAL CONTEXT / S10a + S10b

## Difficult peripheral access

Published ultrasound-access findings differ by workflow and operator. The comparator for VASCOUT should be ultrasound alone, with a defined task and ground truth.

FIELD RESEARCH / UNVERIFIED

## Low perfusion. Limited resources.

Dehydration, shock, motion, gloves and limited resources motivate proposed studies. VASCOUT benefit under these conditions has not been measured.

## The intended contribution

Make anatomy and uncertainty visible for review. The concept does not select an insertion site, prescribe treatment or demonstrate successful IV access.

S10a / SOURCE  S10b / SOURCE  S07 / SOURCE

# Ultrasound first. Added contrast to test.

Visible transillumination, NIR camera imaging and photoacoustic tomography are distinct methods. [S01]

01 / ACQUIRE

## Ultrasound structure

Candidate linear array; coupling and low contact force.

>

02 / COMPARE

## Experimental PA contrast

Test US alone against US + PA under matched conditions.

>

03 / REVIEW

## Anatomy + uncertainty

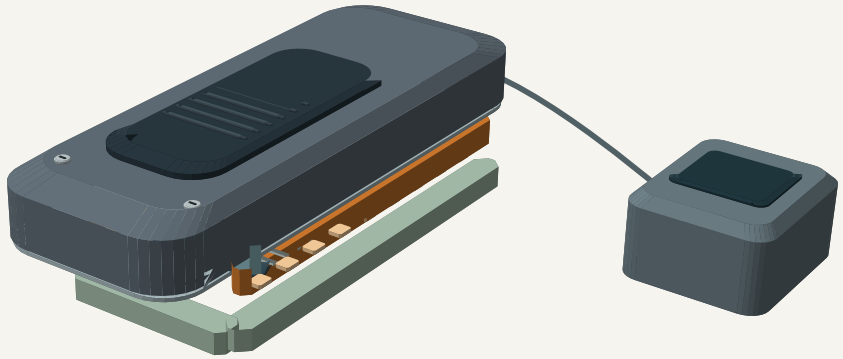
Clinician interprets anatomy and invalid or missing signal.

TARGET: synchronized acquisition, registration and explicit quality feedback. Added value, latency, exposure and thermal budgets remain UNVERIFIED.

S01 / SOURCE S02 / SOURCE S03 / SOURCE

# A probe within a larger research system.

Illustrative geometry from the interactive VASCOUT model. No built prototype or measured packaging performance.



01 / HANDLE

## Deliberate contact

Grip, controls and proposed force feedback need usability studies.

02 / INTERFACE

## Passive contact shoe

Open corridor; acoustic and optical transmission require measurement.

03 / SYSTEM

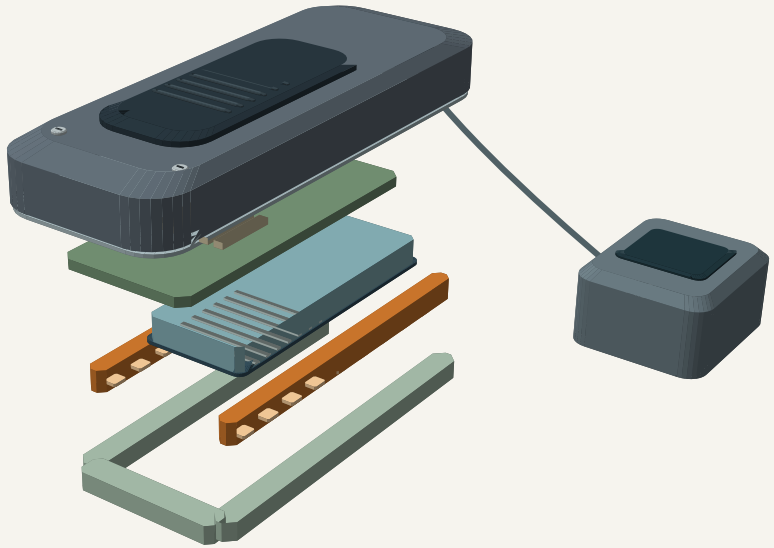
## Separate acquisition unit

The source, acquisition chain and power budget are not contained in the handle.

TARGET / ILLUSTRATIVE / NOT TO SCALE

# Five interfaces to engineer and test.

Layer separation is illustrative; materials, interfaces and integration remain design work.



### 01 Housing

Gloved reach, cleaning, balance and contact-force usability studies.

### 02 Acoustic sensor

Point-spread function, vessel localization, near-field performance and deformation.

### 03 Optical rails

Incremental benefit over US alone, exposure, heating, pigmentation effects and motion.

### 04 Contact shoe

Transmission loss, air gaps, vessel compression, cleaning and sterility requirements.

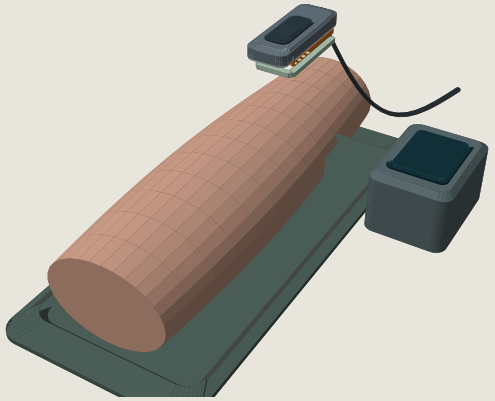
### 05 Electronics

Synchronization, registration, failure states, latency, power and thermal budgets.

SEPARATE SOURCE / ACQUISITION UNIT REQUIRED

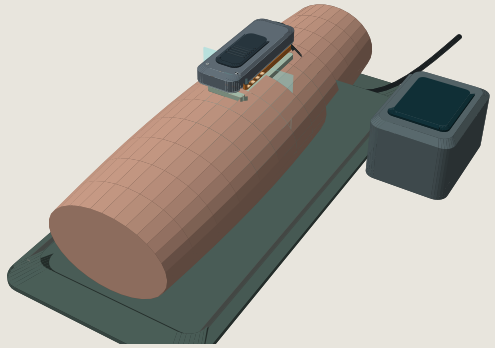
# From a sweep to clinician review.

Static frames from a proposed 30-second workflow. Synthetic anatomy, pre-authored vessels and no measured patient data.



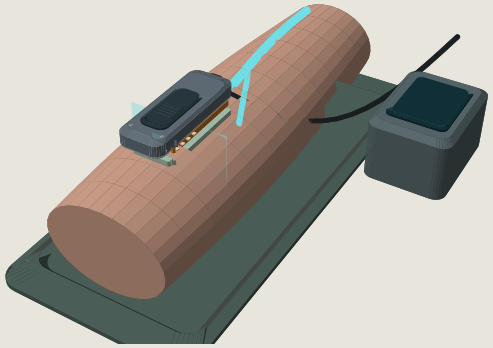
**00s / FIELD SETUP**

Proposed probe, tether and field pad.



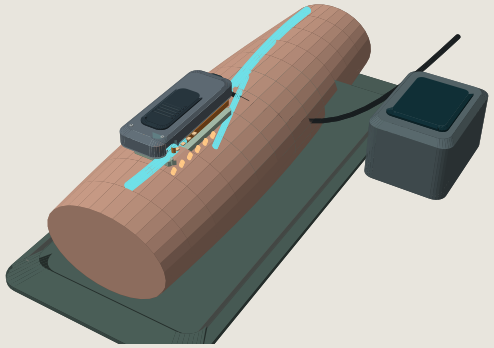
**10s / SIMULATED SCAN**

Drawn scan plane; no ultrasound acquisition.



**19s / VESSEL MAP**

Pre-authored path; no vessel detected.



**23s / CLINICIAN REVIEW**

Unresolved anatomy; no IV outcome.

CONCEPT ONLY: no scan generated, no vessel measured, no insertion guidance and no demonstrated access success.

**Open the live animated concept**

# Numbers are inputs to investigation.

All values below are TARGETS or experiment inputs. They are not demonstrated device specifications.

| | | |
|---|---|---|
| **Probe envelope** | 150 x 56 x 34 mm | Illustrative packaging; ergonomics and integration pending. |
| **Ultrasound band** | 7-15 MHz candidate | Hardware selection and measured imaging response pending. |
| **PA illumination** | 850 nm LED / 1064 nm pulsed | Alternative laboratory configurations; no combined claim. |
| **Phantom depths** | 2 / 5 / 10 / 15 mm | Test positions; not validated operating depths. |
| **Phantom diameters** | 1 / 2 / 4 / 6 mm | Test sizes; not clinical access thresholds. |
| **3D acquisition** | Tracked sweep / bench scan | Registration and motion sensitivity unmeasured. |
| **Contact / invalid data** | Passive shoe + force / UNKNOWN | Deformation, coupling, safety and failure modes to test. |

**NO DEPTH, FRAME-RATE, SAFETY OR FIELD-READY PERFORMANCE CLAIM**

# What VASCOUT must still establish.

Current position: concept design and literature review. Every device-level gate below is UNVERIFIED.

01 / IMAGING / UNVERIFIED

## Resolution, localization and coverage

Benchmark against known geometry across the phantom matrix.

02 / HYBRID BENEFIT / UNVERIFIED

## Incremental benefit over ultrasound

Define a task, comparator, success criteria and added time.

03 / EQUITY + QUALITY / UNVERIFIED

## Pigmentation, motion and signal dropout

Measure failure modes; no skin-tone invariance established. [S06]

04 / CONTACT + SAFETY / UNVERIFIED

## Deformation, exposure and heating

Measure optical/acoustic exposure, force and thermal budgets.

05 / FIELD USABILITY / UNVERIFIED

## Gloves, cleaning and constrained use

Simulated use before any assertion of field suitability.

06 / CLINICAL + REGULATORY / UNVERIFIED

## Intended use and evidence pathway

Expert review and appropriate approvals before human studies. [S11]

S06 / SOURCE

S11 / SOURCE

# A path from concept to evidence.

Advancement depends on measured results and predefined failure criteria; no stage is marked complete.

| 01 | 02 | 03 | 04 | 05 |
|---|---|---|---|---|
| **BENCH COMPARISON** | **CALIBRATED PHANTOMS** | **SIMULATED USE** | **EXPERT REVIEW** | **APPROVED HUMAN STUDIES** |
| Choose components, ground truth, calibration and US-only comparator. | Measure localization, diameter error, dropout, bias and deformation. | Test gloves, motion, lighting, cleaning, setup and failure recognition. | Refine intended use, risk controls and study requirements. | Only after appropriate approvals; evaluate safety and intended benefit. |
| UNVERIFIED | UNVERIFIED | UNVERIFIED | UNVERIFIED | UNVERIFIED |

Decision at each gate: proceed, revise or stop based on agreed evidence. Foreign-body localization remains a separate future phantom study; negative imaging cannot exclude fragments. [S08, S09]

# A first study to scope together.

POCUS workflow and image-quality research / Proposed fit is an INFERENCE. No partner interest confirmed.

### PROPOSED FIRST STUDY / TARGET

With a clinical investigator, define a simulated vascular-assessment task comparing ordinary ultrasound review with explicit image-quality/uncertainty feedback. Treat any future AI component as a separately validated research module.

### PROPOSED OUTPUT / TARGET

A workflow endpoint set, annotation/quality rubric and investigator-led protocol suitable for the program's scope.

### VALIDATION + ENGAGEMENT GAPS / UNVERIFIED

Direct page fetches returned redirect loops; indexed official text supports the lead. Current intake availability, PI eligibility, equipment/support and PA fit need confirmation. No support or funding is promised.

Explore the device and evidence library

DRAFT RESEARCH DISCUSSION / NOT SENT

# FUJIFILM Sonosite

POCUS workflow and image-quality research / Proposed research fit is an INFERENCE. Partner interest is unconfirmed.

### SOURCED CAPABILITY / P05, P06

Sonosite's official indexed pages describe military POCUS work and invite clinician/researcher collaboration on POCUS AI, validation, user testing and workflows under research agreements.

### PROPOSED FIRST STUDY / INFERENCE

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### PUBLISHED ROUTE / NO OUTREACH SENT

Published research-collaboration page; first confirm current program scope and clinician eligibility.

P05 / OFFICIAL SOURCE

P06 / OFFICIAL SOURCE

### TAILORED DRAFT PITCH / PREPARED WITH AI / NOT SENT

VASCOUT is an unvalidated concept for making peripheral anatomy and imaging uncertainty clearer to clinicians. Your published POCUS collaboration route suggests a possible fit for a clinician-led quality-feedback study, rather than an assumption of PA hardware support. Would your research team discuss eligibility and scope for a simulated vascular-assessment protocol comparing ultrasound review with explicit poor-signal feedback? A clinical investigator would define the task and endpoints; any AI component would require separate validation. We seek protocol and workflow feedback first, with resources subject to a formal agreement.

**Official engagement route**

CLINICAL BENEFIT / SAFETY / FIELD SUITABILITY: UNVERIFIED

# Sources / clinical and imaging context

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S01 / Current DFU / Manufacturer

### Veinlite LEDX - directions for use

Manufacturer DFU. Visible transillumination; not NIR or PA tomography. No comparative efficacy proof.

S02 / 2024 / Healthy human

### Photoacoustic vector tomography for deep haemodynamic imaging

Reported healthy-human vascular imaging. Different research system; full-text access incomplete.

S03 / 2021 / Healthy human

### Photoacoustic imaging of human vasculature using LED versus laser illumination

LED/laser research comparison in phantoms and healthy limbs. No handheld field validation; publisher access limited.

S04 / 2022 / Clinical

### Monitoring peripheral hemodynamic response to changes in blood pressure via photoacoustic imaging

Dialysis/exercise hemodynamics. Not a dehydration meter or shock classifier; root full-text reread incomplete.

S05 / 2022 / Phantom

### Thin ceramic PZT dual- and multi-frequency pMUT arrays for photoacoustic imaging

Sensor-array prior art in agar. No real-time human access proof; direct PMC access challenged.

S06 / 2026 / Healthy human

### The confounding effects of skin colour in photoacoustic imaging

42 healthy participants; optical/acoustic confounding. No demonstrated VASCOUT skin-tone invariance.

S10a / 2005 / Clinical

### Ultrasonography-guided peripheral intravenous access versus traditional approaches

60 difficult-access patients; reported benefit. Odd/even-day allocation, two operators; full text unverified. Read with S10b.

S10b / 2009 / Clinical

### Ultrasonographically guided peripheral intravenous cannulation: a randomized trial

59-patient randomized ED trial; no demonstrated superiority. Workflow-specific; full text unverified. Read with S10a.

# Sources / limitations, engineering and prior art

Research snapshot: 20 October 2026. URLs link to the retained source record; source evidence does not validate VASCOUT.

S07 / 2026 / Guideline

### TCCC Guidelines - vascular access claim audit

1 May 2026 guideline, directly read in the retained audit. Clinical context only; no treatment directions in this deck.

S08 / 2009 / Cadaver

### Bedside ultrasound for detection of soft tissue foreign bodies: a cadaveric study

Cadaver study; small objects and poor reliability. Negative imaging cannot universally exclude fragments.

S09 / 2018 / Simulation

### Ultrasound evaluation of soft-tissue foreign bodies by US Army medics

28 medics, chicken-thigh simulation after training. No injured-human or battlefield validation; retrieval limits retained.

S11 / 2021 / Regulator

### FDA: Imagio Breast Imaging System - P200003

FDA approval for another product and breast-imaging indication. No VASCOUT clearance or pathway follows.

S12 / 2018 / Healthy human

### A device for improving the visual clarity and dimension of veins

20 healthy volunteers in vacuum-device research. Not shock evidence; retrieval gaps. Suction excluded here.

L01 / Manufacturer / Manufacturer

### TI AFE5832LP - ultrasound analog front end

Manufacturer: 32 receive inputs, 16 multiplexed ADC channels. Complete system architecture unresolved.

L02 / Manufacturer / Manufacturer

### Analog Devices MAX2082 - octal ultrasound transceiver

Manufacturer: octal transmit/receive device. Does not prove proposed system integration or power budget.

G01 / 2026 / Concept material

### User-supplied Gemini concepts - audited research inputs

User-supplied Gemini proposals, not studies. Audited concept inputs; no scientific validation. See live evidence library.

# Sources / this research prospect

Accessed 20 October 2026 EDT / 21 October UTC. Organization capability and route evidence; not VASCOUT validation.

P05

## FUJIFILM Sonosite - Collaborate With Us on Advancing POCUS AI

Official indexed page text read; direct open returned redirect loop

Official indexed page invites clinicians/researchers for POCUS AI validation, workflow and user testing; support is conditional and proposal-specific.

LIMIT: Availability needs reconfirmation. Program is AI-focused and clinician-led; no commitment to PA hardware or this concept. Eligibility, support and funding unverified.

P06

## FUJIFILM Sonosite - Military Medicine Ultrasound

Official indexed page text read; direct open returned redirect loop

Official indexed page describes military point-of-care ultrasound work.

LIMIT: Historical/company-reported context; not endorsement, acceptance or VASCOUT field evidence.

PUBLIC SOURCES / PROPOSED FIT IS INFERENCE / INTEREST UNCONFIRMED